

Neo4j Full Questions and Answers

Question 1: Social Network Graph - People and Friends

// Create 5 person nodes

```
CREATE (:Person {id: 1, name: 'Alice', age: 24, location: 'Bangalore'}),  
      (:Person {id: 2, name: 'Bob', age: 28, location: 'Delhi'}),  
      (:Person {id: 3, name: 'Charlie', age: 22, location: 'Bangalore'}),  
      (:Person {id: 4, name: 'Diana', age: 26, location: 'Mumbai'}),  
      (:Person {id: 5, name: 'Eva', age: 23, location: 'Chennai'});
```

// Create FRIEND relationships

```
MATCH (a:Person), (b:Person)  
WHERE a.id < b.id  
CREATE (a)-[:FRIEND]->(b);
```

// Add city property using SET

```
MATCH (p:Person)  
SET p.city = p.location;
```

// Remove city property for people in Bangalore

```
MATCH (p:Person)  
WHERE p.city = 'Bangalore'  
REMOVE p.city;
```

// Delete persons older than 25

```
MATCH (p:Person)  
WHERE p.age > 25
```

```
DELETE p;
```

Question 2: Product Catalog - Category, Product, and Search

```
// Create category nodes
```

```
CREATE (:Category {name: 'Electronics'}),  
       (:Category {name: 'Books'}),  
       (:Category {name: 'Fashion'}),  
       (:Category {name: 'Techs'}),  
       (:Category {name: 'Toys'});
```

```
// Create product nodes and BELONGS_TO relationships using FOREACH
```

```
WITH ['Laptop', 'T-shirt', 'Teddy', 'Television', 'Tablet'] AS products
```

```
FOREACH (p IN products |  
  CREATE (prod:Product {name: p, price: toInteger(rand() * 1000)})  
);
```

```
// Link each product to a random category (example)
```

```
MATCH (prod:Product), (cat:Category)  
WHERE cat.name IN ['Electronics', 'Toys', 'Fashion', 'Techs']  
WITH prod, cat LIMIT 5  
MERGE (prod)-[:BELONGS_TO]->(cat);
```

```
// Find products with "T" in their name
```

```
MATCH (p:Product)  
WHERE p.name CONTAINS 'T'  
RETURN p.name, p.price;
```

```
// Return products sorted by price (ascending)
```

```
MATCH (p:Product)
```

```
RETURN p.name, p.price
```

```
ORDER BY p.price ASC;
```

Question 3: Movie Graph - Actors and Films

```
// Create actor and movie nodes
```

```
CREATE (:Actor {name: 'Tom'}),
```

```
      (:Actor {name: 'Sara'}),
```

```
      (:Actor {name: 'Mike'}),
```

```
      (:Actor {name: 'Lara'}),
```

```
      (:Actor {name: 'John'}),
```

```
      (:Movie {title: 'Fast Time'}),
```

```
      (:Movie {title: 'Gone'}),
```

```
      (:Movie {title: 'Fury'}),
```

```
      (:Movie {title: 'Drama King'}),
```

```
      (:Movie {title: 'Flashback'});
```

```
// Connect actors to movies
```

```
MATCH (a:Actor), (m:Movie)
```

```
WITH a, m LIMIT 10
```

```
MERGE (a)-[:ACTED_IN]->(m);
```

```
// Find movies starting with "F"
```

```
MATCH (m:Movie)
```

```
WHERE m.title STARTS WITH 'F'
```

```
RETURN m.title;
```

```
// Count movies per actor

MATCH (a:Actor)-[:ACTED_IN]->(m:Movie)

RETURN a.name, COUNT(m) AS movie_count;
```

```
// Top actors by number of movies

MATCH (a:Actor)-[:ACTED_IN]->(m:Movie)

RETURN a.name, COUNT(m) AS movie_count

ORDER BY movie_count DESC

LIMIT 3;
```

Question 4: Event Attendance System

```
// Create Event and User nodes
```

```
CREATE (:Event {name: 'Tech-Con'}),

      (:Event {name: 'Dev-Summit'}),

      (:User {name: 'John'}),

      (:User {name: 'Sara'});
```

```
// Link users to events with ATTENDED relationships and set a rating
```

```
MATCH (u:User), (e:Event)

WITH u, e LIMIT 4

MERGE (u)-[r:ATTENDED]->(e)

SET r.rating = toInteger(rand() * 5);
```

```
// Remove John's attendance
```

```
MATCH (u:User {name: 'John'})-[r:ATTENDED]->(e:Event)

DELETE r;
```

```
// Remove rating < 3
```

```
MATCH (:User)-[r:ATTENDED]->(:Event)
```

```
WHERE r.rating < 3
```

```
REMOVE r.rating;
```

```
// Calculate total rating given by Sara
```

```
MATCH (:User {name: 'Sara'})-[r:ATTENDED]->(:Event)
```

```
RETURN SUM(r.rating) AS total_rating;
```

Question 5: City Import and Clean-up

```
// Create city nodes using FOREACH
```

```
WITH ['Berlin', 'Delhi', 'Madrid', 'Rome', 'Osaka'] AS cities
```

```
FOREACH (city IN cities |
```

```
    CREATE (:City {name: city})
```

```
);
```

```
// Add region property
```

```
MATCH (c:City)
```

```
SET c.region = CASE c.name
```

```
    WHEN 'Delhi' THEN 'Asia'
```

```
    WHEN 'Osaka' THEN 'Asia'
```

```
    ELSE 'Europe'
```

```
END;
```

```
// Find cities starting with "D"
```

```
MATCH (c:City)
```

```
WHERE c.name STARTS WITH 'D'
```

```
RETURN c.name;
```

```
// Delete cities in Europe
```

```
MATCH (c:City)
```

```
WHERE c.region = 'Europe'
```

```
DELETE c;
```

```
// Count remaining cities
```

```
MATCH (c:City)
```

```
RETURN COUNT(c) AS city_count;
```